DATABASE NORMALIZATION

Corrected Study Guide for RAG Retrieval

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Purpose: a structured study guide designed for topic-wise learning and retrieval practice.

1. What Is Database Normalization

Database normalization is the process of organizing relational tables to reduce avoidable duplication and prevent insertion, update, and deletion anomalies. It uses keys and functional dependencies to decide how attributes should be grouped.

Why normalize?

• Store each fact in an appropriate place.

• Reduce inconsistent copies of the same fact.

• Make updates safer and data relationships clearer.

• Normalization improves design, but joins may be needed to reconstruct information.

2. Functional Dependencies

A functional dependency X → Y means that each value of attribute set X determines exactly one value of attribute set Y. X is the determinant. A candidate key is a minimal set of attributes that functionally determines every attribute in a relation.

Example

```
StudentID → StudentName
CourseID → CourseName
(StudentID, CourseID) → Grade
```

A partial dependency occurs when a non-key attribute depends on only part of a composite candidate key. A transitive dependency occurs when a key determines a non-key attribute that in turn determines another non-key attribute.

3. Anomalies in Poorly Designed Tables

Insertion anomaly

A fact cannot be stored without inserting an unrelated fact. Example: a new course cannot be recorded until a student enrolls.

Update anomaly

The same fact appears in multiple rows and must be changed everywhere. Missing one row creates inconsistent data.

Deletion anomaly

Deleting one row accidentally removes another fact. Deleting the last enrolment might also erase the only record of a course.

4. First Normal Form (1NF)

A relation is in First Normal Form when each cell contains a single atomic value for the chosen data model, and there are no repeating groups or multi-valued columns.

Before and after

```
Not 1NF:
StudentID | Name | PhoneNumbers
1         | Asha | 9876, 9123

1NF design:
StudentID | Name | Phone
1         | Asha | 9876
1         | Asha | 9123
```

Do not treat 'atomic' as a universal property of a value. It depends on how the application needs to query and manage the data.

5. Second Normal Form (2NF)

A relation is in 2NF when it is in 1NF and every non-prime attribute is fully functionally dependent on every candidate key. In the common classroom case, this means removing partial dependencies on part of a composite key.

```
Enrollment(StudentID, CourseID, StudentName, CourseName, Grade)
Key: (StudentID, CourseID)
StudentID → StudentName
CourseID → CourseName
(StudentID, CourseID) → Grade
```

Decompose into Student(StudentID, StudentName), Course(CourseID, CourseName), and Enrollment(StudentID, CourseID, Grade).

6. Third Normal Form (3NF)

A relation is in 3NF if, for every non-trivial functional dependency $X \rightarrow A$, either X is a superkey or A is a prime attribute (part of at least one candidate key). A useful introductory rule is to remove transitive dependencies of non-key attributes on a key.

```
Student(StudentID, StudentName, DeptID, DeptName)
StudentID → DeptID
DeptID → DeptName

Decompose:
Student(StudentID, StudentName, DeptID)
Department(DeptID, DeptName)
```

The formal definition is more precise than the shortcut 'no transitive dependency', especially when multiple candidate keys exist.

7. Boyce-Codd Normal Form (BCNF)

A relation is in BCNF if every determinant of every non-trivial functional dependency is a superkey. BCNF is stricter than 3NF.

A 3NF relation can fail BCNF when a dependency has a determinant that is not a superkey, even though the dependent attribute is prime. Decomposing to BCNF can sometimes sacrifice dependency preservation, so design decisions may involve trade-offs.

8. Comparison at a Glance

Form	Main requirement	Common issue addressed
1NF	Atomic values; no repeating groups	Multi-valued cells
2NF	1NF + no partial dependency of non-prime attributes	Composite-key partial dependency
3NF	For each $X \rightarrow A$, X is superkey or A is prime	Certain transitive dependencies
BCNF	Every determinant is a superkey	Determinants that are not keys

9. Worked Normalization Example

Consider the relation below. Assume StudentID identifies a student, CourseID identifies a course, and the pair identifies an enrolment.

```
Enrollment(StudentID, StudentName, CourseID, CourseName, Instructor, Grade)
Key: (StudentID, CourseID)
StudentID → StudentName
CourseID → CourseName, Instructor
(StudentID, CourseID) → Grade
```

A normalized decomposition is Student(StudentID, StudentName), Course(CourseID, CourseName, Instructor), and Enrollment(StudentID, CourseID, Grade). Check the assumptions: if instructors vary by course offering or semester, the key and dependencies must change.

Validation checklist

• Identify all candidate keys.

• List functional dependencies from the business rules.

• Check atomic values and repeating groups.

• Remove partial dependencies for 2NF.

• Check 3NF and BCNF using the formal definitions.

• Verify lossless joins and consider dependency preservation.

10. Common Mistakes

• Assuming every table with a primary key is automatically normalized.

• Confusing a superkey with a candidate key: a candidate key is minimal.

• Applying 2NF reasoning without checking whether the key is composite.

• Treating 3NF and BCNF as identical.

• Decomposing tables without checking whether joins lose information.

• Inventing functional dependencies without business-rule evidence.

11. Quick Revision and Practice

Quick revision

• 1NF: atomic values and no repeating groups.

• 2NF: 1NF plus no partial dependency of non-prime attributes on a candidate key.

• 3NF: every non-trivial $X \rightarrow A$ has X as a superkey or A as a prime attribute.

• BCNF: every determinant of a non-trivial dependency is a superkey.

Practice questions

• A table stores three phone numbers in one column. Which normal form is violated, and why?

• For key (StudentID, CourseID), StudentID → StudentName. What dependency problem appears?

• Given EmployeeID → DeptID and DeptID → DeptName, explain the likely 3NF decomposition.

• State the formal difference between 3NF and BCNF.

• Why must a decomposition be checked for lossless join?